MIL-D-51106A(EA)
26 May 1982
SUPERSEDING
MIL-D-51106(MU)
11 January 1963

MILITARY SPECIFICATION

N,N'-DICYCLOHEXYLCARBODIIMIDE, TECHNICAL

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of N,N'dicyclohexylcarbodiimide.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D1193 - Reagent Water
D1493 - Solidification Point of Industrial Organic Chemicals
E443 - Sulfur in Organic Compounds by Oxygen Flask Combustion

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Chemical and physical characteristics. The N,N'dicyclohexylcarbodiimide shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Minimum	Maximum	Test paragraph
Purity, percent by weight	95.0	---	4.2.4.1
Sulfur content, percent by weight	---	0.1	4.2.4.2
Solidification point, °C	34	---	4.2.4.3
Total solids, percent by weight	---	0.5	4.2.4.4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the N,N'dicyclohexylcarbodiimide produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test (see 6.2). Sampling shall be conducted in accordance with table II. A representative specimen of approximately 200 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

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TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container ready for shipment. Sample shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Protective interior drum coating missing or not as specified
(g) Gaskets for closures missing or not as specified
(h) Palletization missing or not as specified (when required)

4.2.3.2 For test (see 6.2). Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 400 g. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests (see 6.2). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay.

(a) 1,4-Dioxane. Reflux the 1,4-dioxane over calcium hydride and distill it to 0.05 percent or less water. Store in a desiccator.

(b) Oxalic acid solution. Dry oxalic acid in vacuo at 50°C for 4 hours, breaking the crust every hour. Prepare a 0.1N solution using 1,4-dioxane as the solvent.

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(c) Procedure. Weigh to the nearest 0.1 milligrams 0.070 to 0.150 g of the specimen in a micro weighing bottle or glass ampoule. Place the bottle or ampoule in a dry 250-milliliter (ml) standard taper Erlenmeyer flask containing 40 ml of the oxalic acid solution. If the container is a glass ampoule, crush it with a glass rod to release the contents and wash down the rod with 5 ml of dry dioxane; if a weighing bottle is used, remove the cover as soon as the bottle has been placed in the oxalic acid solution, and wash down the implement used in removing the cover with 5 ml of dry dioxane. Insert a water-cooled condenser (equipped with standard taper drip joint at bottom) into the flask and reflux the contents of the flask for 15 minutes on a hot plate. Wash down the condenser with 10 to 15 ml of water. Remove the flask from the hot plate, cool, add 100 ml of water, and titrate with standardized 0.1N sodium hydroxide solution to the first pink using 1 ml of a 1 percent solution of phenolphthalein in ethanol as the indicator. Run a blank under the same conditions on 40 ml of the oxalic acid solution. Calculate the percent N,N'-dicyclohexylcarbodiimide as follows:

$$\text{Percent N,N'dicyclohexylcarbodiimide} = \frac{10.32(A-B)C}{W}$$

where: A = Milliliters of sodium hydroxide required to titrate the blank,
B = Milliliters of sodium hydroxide required to titrate the specimen,
C = Normality of the sodium hydroxide solution, and
W = Grams of specimen.

4.2.4.2 Sulfur. Determine the sulfur content of the specimen in accordance with ASTM E443.

4.2.4.3 Solidification point. Determine the solidification point of the specimen in accordance with ASTM D1493.

4.2.4.4 Solids. Heat 100 g of the specimen weighed to the nearest 0.1 g, in a beaker to between 50° and 55°C. Filter through a tared Gooch crucible, containing a suitable mat. Wash the residue with 100 ml of carbon tetrachloride at 25°C. Dry the crucible and residue at 100°C for one hour, cool to room temperature, and weigh. Calculate the percent solids as follows:

$$\text{Percent solids} = \frac{100A}{B}$$

where: A = Grams of residue, and
B = Grams of specimen.

5. PACKAGING

5.1 Unit packing. N,N'-Dicyclohexylcarbodiimide shall be unit packed level A or C, as specified (see 6.2).

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5.1.1 Level A. A specified quantity (see 6.2) of N,N'-dicyclohexylcarbodiimide shall be unit packed in a nominal 55-gallon (gal) capacity steel drum conforming to Department of Transportation (DOT) specification 5, 5A or 5B. Each drum shall be furnished with a protective interior drum coating, and gaskets for closures which shall neither affect nor be affected by the contents. The drum shall have a 3/4-inch threaded plug-flange vent and a 2-inch threaded plug-flange opening. These shall be gasketed and torqued closed in accordance with the drum manufacturer's recommendations.

5.1.2 Level C. A specified quantity (see 6.2) of N,N'-dicyclohexylcarbodiimide shall be unit packed level C in a manner to assure retention of the specified quantity and quantity of contents, and the integrity of the container. The container shall conform to DOT regulations, Uniform Freight Classification Rules, National Motor Freight Classification Rules, and other carrier rules applicable to the mode of transportation.

5.2 Packing. The N,N'-dicyclohexylcarbodiimide unit packed as specified in 5.1 shall require no further protection for shipment.

5.3 Unitization. Shipping containers, when specified (see 6.2), shall be palletized in accordance with MIL-STD-147.

5.4 Marking. Containers shall be marked in accordance with MIL-STD-129, and shall show the date of manufacture, the lot or batch number of the N,N'dicyclohexylcarbodiimide and precautionary marking.

6. NO 'ES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of unit packing required (see 5.1).
(c) Unit quantity required (see 5.1.1 and 5.1.2).

6.2 Sampling and testing precautions. This material is dangerous when in contact with skin and eyes. It is known to cause burns and dermatitis. In case of contact with skin or eyes, immediately flush with water for at least 15 minutes; for eyes get medical attention. All people handling this material should be aware of its dangerous nature.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding off method of ASTM E29.

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